

Chapter 1

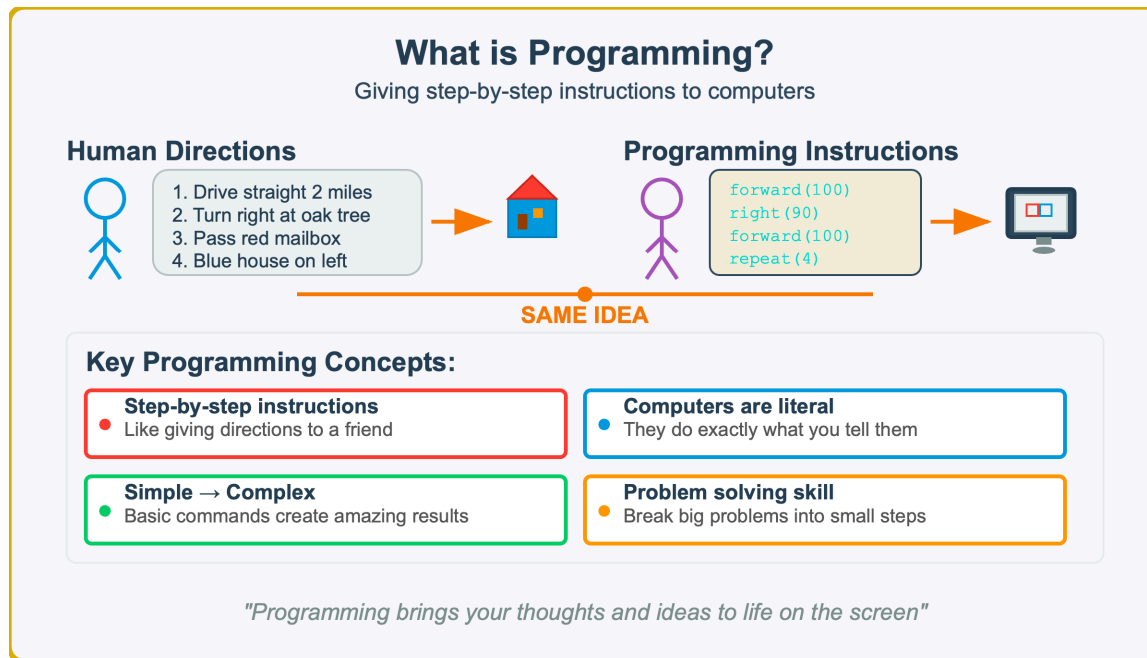


Figure 1.1 Programming as Step—by—Step Instructions — A comparison between giving human directions (left) and writing computer programs (right), illustrating that programming is fundamentally about providing clear, sequential instructions to solve problems.

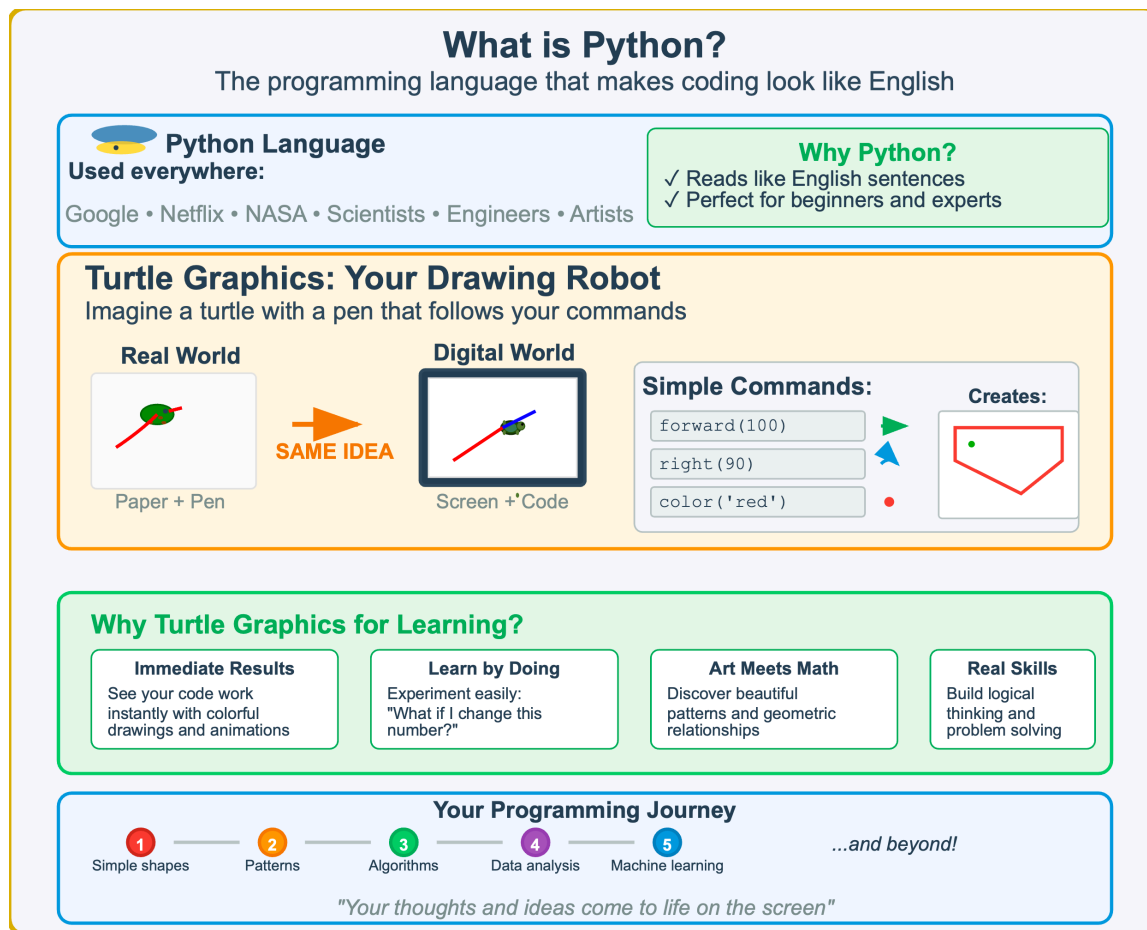


Figure 1.2 — Learning Programming with Python Turtle Graphics — Comparing real—world turtle drawing with digital turtle graphics, showing how simple commands like forward(100) and right(90) create visual results while building fundamental programming skills.

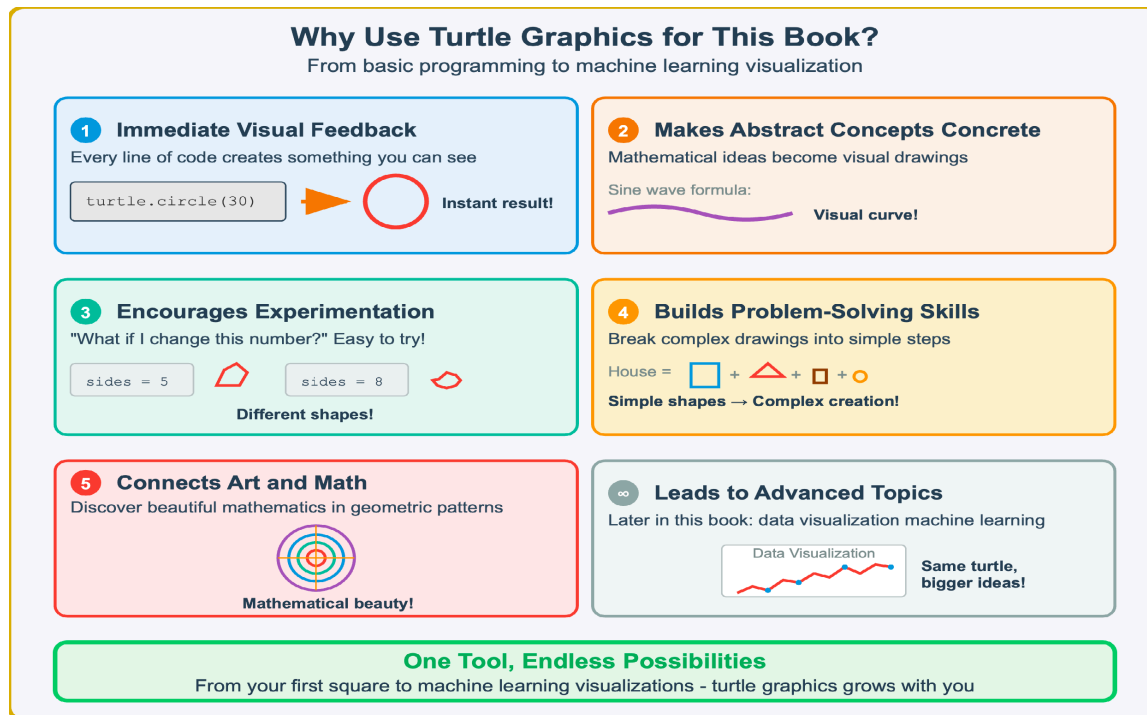


Figure 1.3 — The Educational Value of Turtle Graphics — From basic shape drawing to advanced data visualization and machine learning, turtle graphics provides a continuous learning pathway that builds fundamental programming skills while maintaining visual engagement.

Setting Up Your Programming Tools

Option 1: Thonny - Your Desktop Programming Companion

T **Thonny**
Install on your computer

R **Replit**
Runs in web browser

👉 We'll focus on Thonny first

Thonny Interface Preview

Thonny

Editor (Write your code here):

```
import turtle
my_turtle = turtle.Turtle()
my_turtle.circle(50)
```

Shell (See results here):

```
>>> Running your program...
Circle drawn successfully!
```

How to use:
1. Type code in editor 2. Press F5 to run 3. See results in shell

Why Choose Thonny?

Beginner-Friendly
Simple, clean interface
No overwhelming features

Built-in Python
Python included
No separate install

Step Debugging
Watch code run
line by line

Works Offline
No internet needed
once installed

Completely Free
Open source
Always free to use

Made for Learning
Designed for
beginners

Perfect for This Book!

Ready to Start? Download from thonny.org → Install → Start coding your first turtle graphics!
Next: We'll walk through the installation process step-by-step

Figure 1.4 — Introduction to Thonny — A Beginner—Friendly Python Development Environment — Overview of Thonny’s interface and key features that make it ideal for learning Python programming, including its simple design, built—in Python interpreter, step debugging capabilities, and offline functionality.

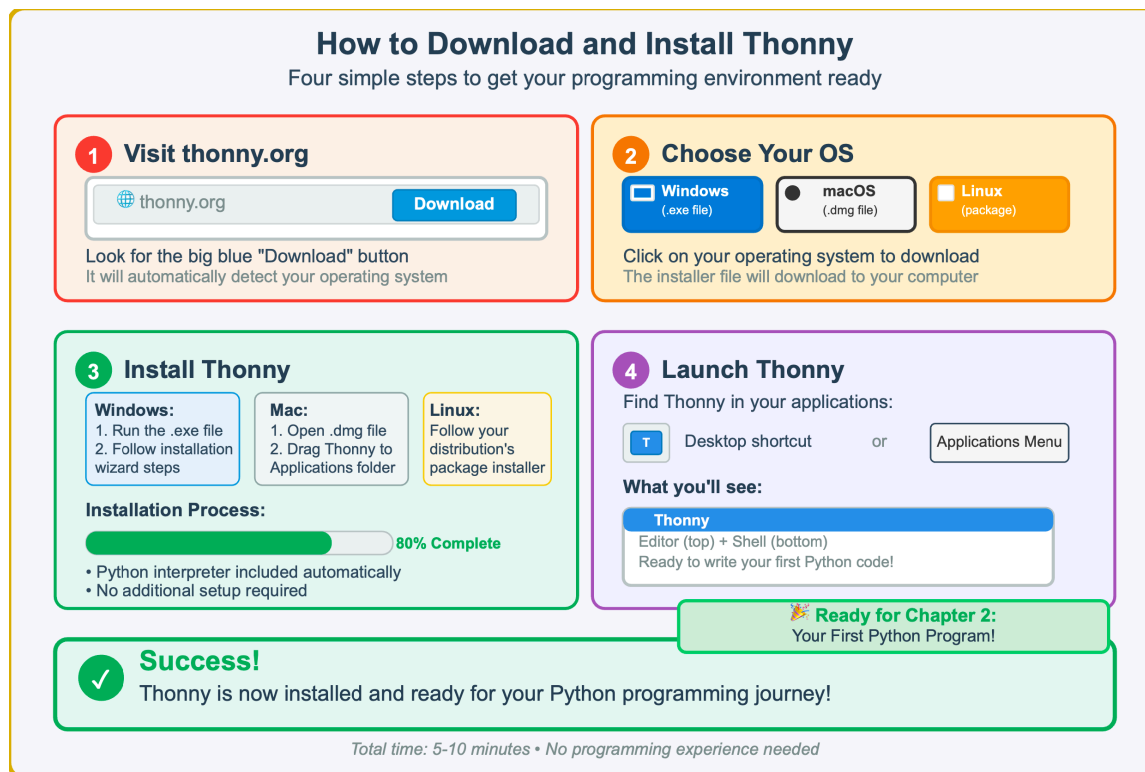


Figure 1.5 — Thonny Installation Guide — A step—by—step walkthrough for downloading and installing Thonny across different operating systems (Windows, macOS, and Linux), from visiting thonny.org to launching your first programming environment.

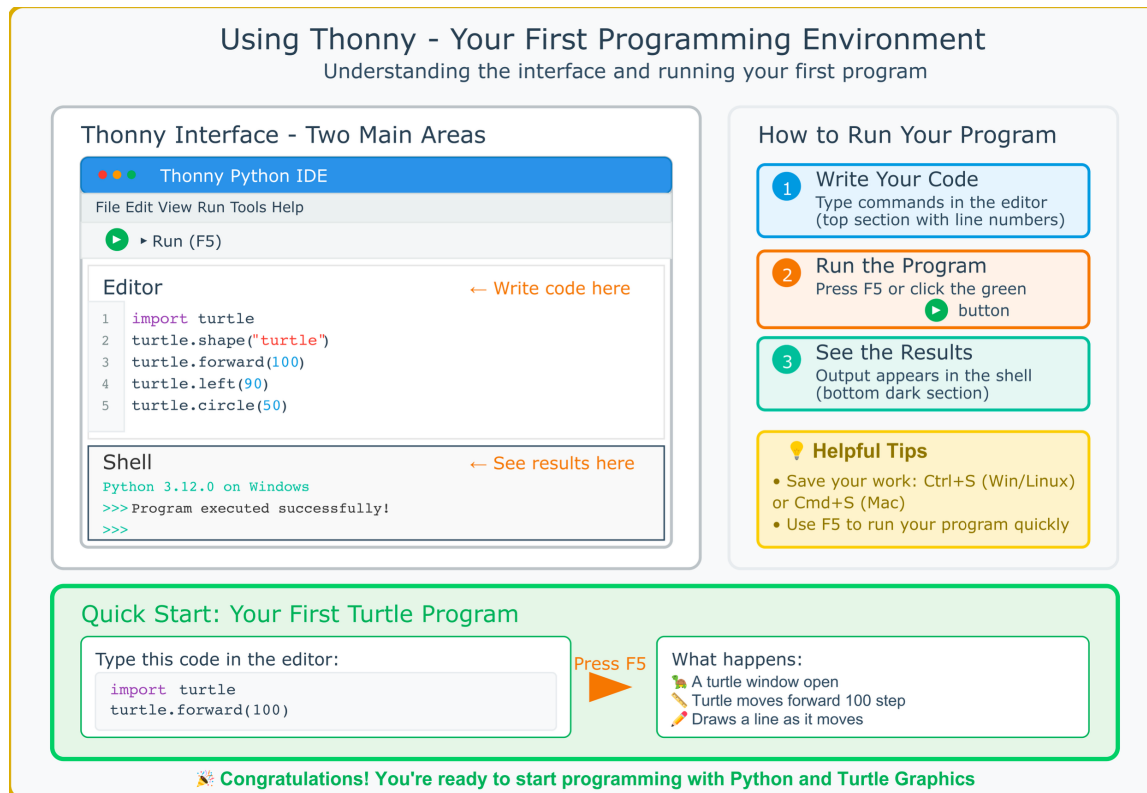
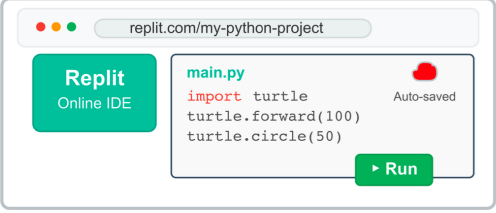


Figure 1.6 — Getting Started with Thonny — Understanding the two—panel interface (editor for writing code, shell for viewing results) and following the simple process of typing code, pressing F5 to run, and seeing your turtle graphics come to life.

Option 2: Replit - Programming in Your Browser

Complete programming setup without installing anything on your computer

Replit Interface in Your Browser



The screenshot shows the Replit online IDE interface. At the top, there's a browser address bar with 'replit.com/my-python-project'. Below it, on the left, is a green 'Replit Online IDE' button. In the center, a code editor shows a file named 'main.py' with the following Python code:

```
import turtle
turtle.forward(100)
turtle.circle(50)
```

 To the right of the code is an 'Auto-saved' status and a red cloud icon. At the bottom right of the code editor is a green 'Run' button.

Why Choose Replit?

- No Installation**
Works in any browser
Start coding instantly
- Access Anywhere**
Cloud-saved projects
Any device, anywhere
- Easy Sharing**
Share projects instantly
Send link to anyone
- Collaboration**
Code together live
Perfect for teams

Getting Started with Replit

Quick Setup (2 minutes):

1. Go to replit.com
2. Sign up (free account)
3. Create new Python project → Start coding!

Perfect for:

- Chromebooks shared computers
- Classroom settings quick projects
- When you want to start immediately

Both Thonny and Replit are excellent choices - pick what works best for your situation!

Figure 1.7 — Replit Online Programming Environment. Replit offers a browser—based Python development environment with no installation required, making it ideal for Chromebooks, shared computers, and situations where you want to start coding immediately. Projects are automatically saved to the cloud and can be accessed from any device.

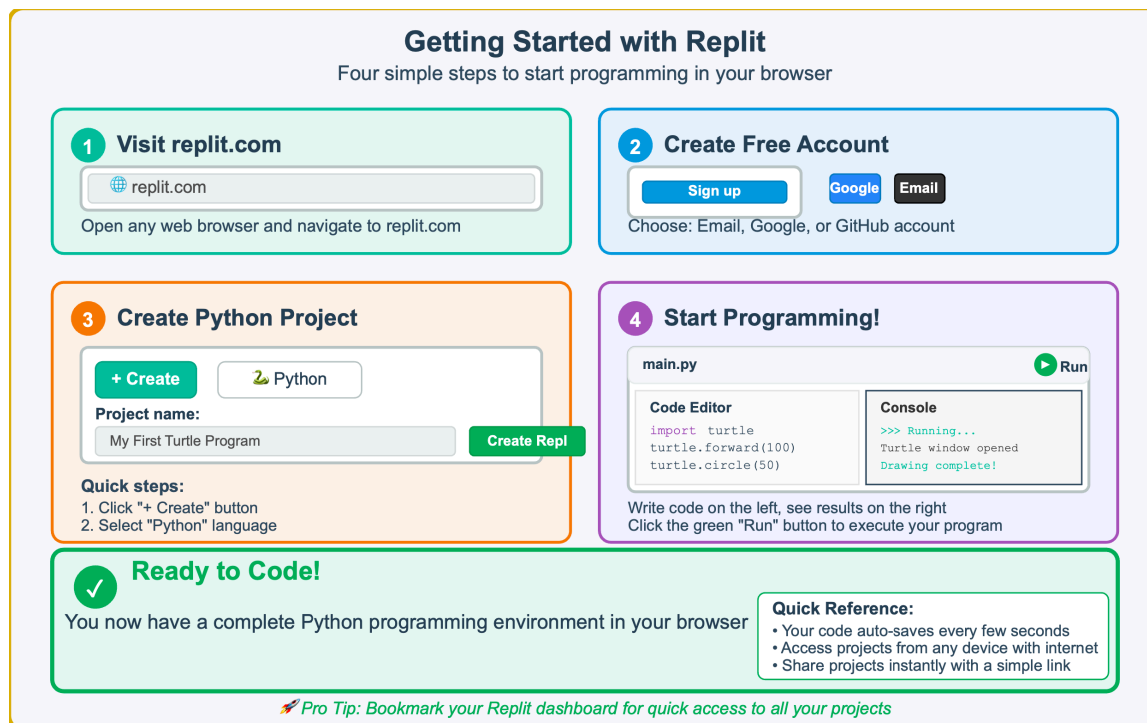


Figure 1.8 — Getting Started with Replit Step—by—Step. This guide shows the complete process from visiting replit.com to running your first Python program, requiring only a web browser and internet connection.

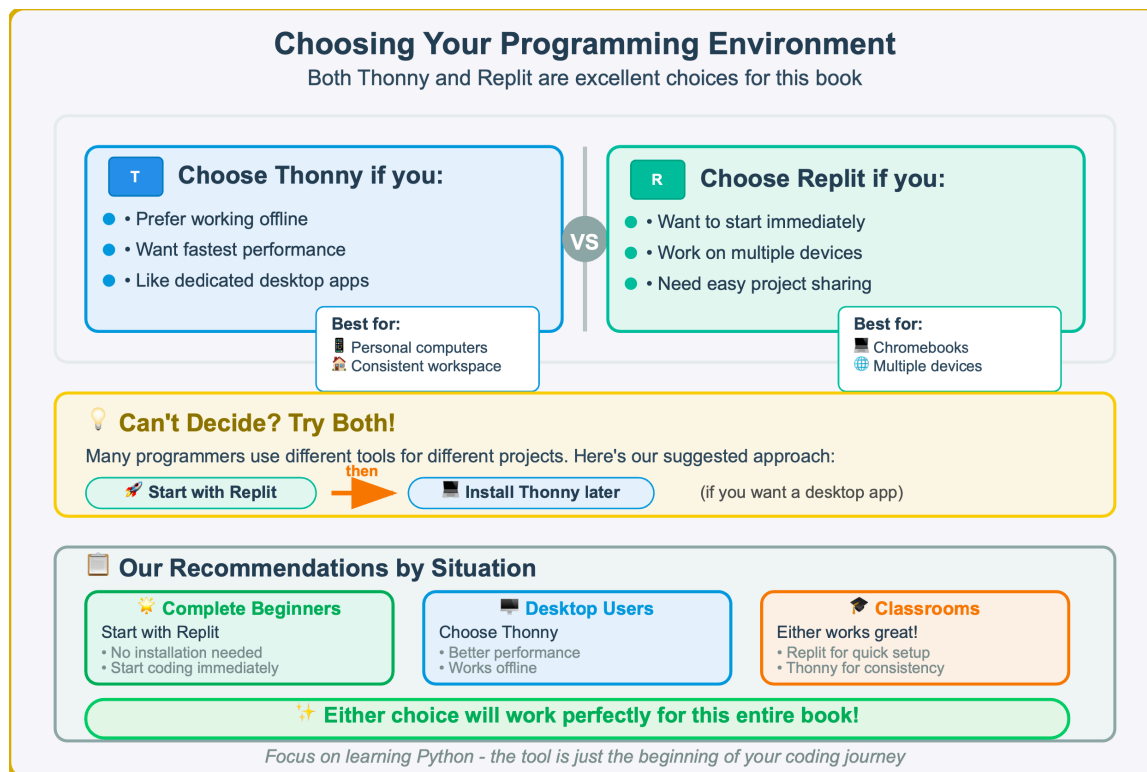


Figure 1.9 — Choosing Your Programming Environment — Thonny vs Replit. This comparison guide helps you select the best programming tool based on your specific needs and situation. Both environments work excellently for all concepts covered in this book.

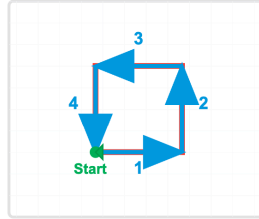
Your First Python Program: Drawing a Square

Don't worry about understanding every detail - we'll explain as we go!

The Complete Code

```
1 import turtle
2 # Create a turtle
3 my_turtle = turtle.Turtle()
4 # Draw a square
5 my_turtle.forward(100)
6 my_turtle.right(90)
7 my_turtle.forward(100)
8 my_turtle.right(90)
9 my_turtle.forward(100)
10 my_turtle.right(90)
11 my_turtle.forward(100)
12 my_turtle.right(90)
13 my_turtle.forward(100)
14 my_turtle.right(90)
15 # Keep window open
16 turtle.done()
17
```

What Your Turtle Draws



Step by Step:

1. Move forward 100 →
 2. Turn right 90°
Move forward 100 ↑
 3. Turn right 90°
Move forward 100 ←
 4. Turn right 90°
Move forward 100 ↓
- Perfect square!**

Key Programming Concepts in This Code

- **import turtle**
Brings in the turtle graphics tools
- **my_turtle = turtle.Turtle()**
Creates your drawing turtle
- **forward() and right()**
Basic movement commands
- **# comments**
Explain what the code does



Try It Yourself!

Copy this code into your programming environment:

1. Open Thonny or Replit
2. Type (or copy) the code exactly as shown

Then run your program:

3. Press F5 (Thonny) or click Run (Replit)
4. Watch your turtle draw a perfect square!

Congratulations - you're now a Python programmer! 🎉

Figure 1.10 — Your First Python Program — Drawing a Square with Turtle Graphics. This complete program demonstrates fundamental programming concepts including importing modules, creating objects, using methods, and controlling program flow. The visual output shows how simple commands combine to create precise geometric shapes.

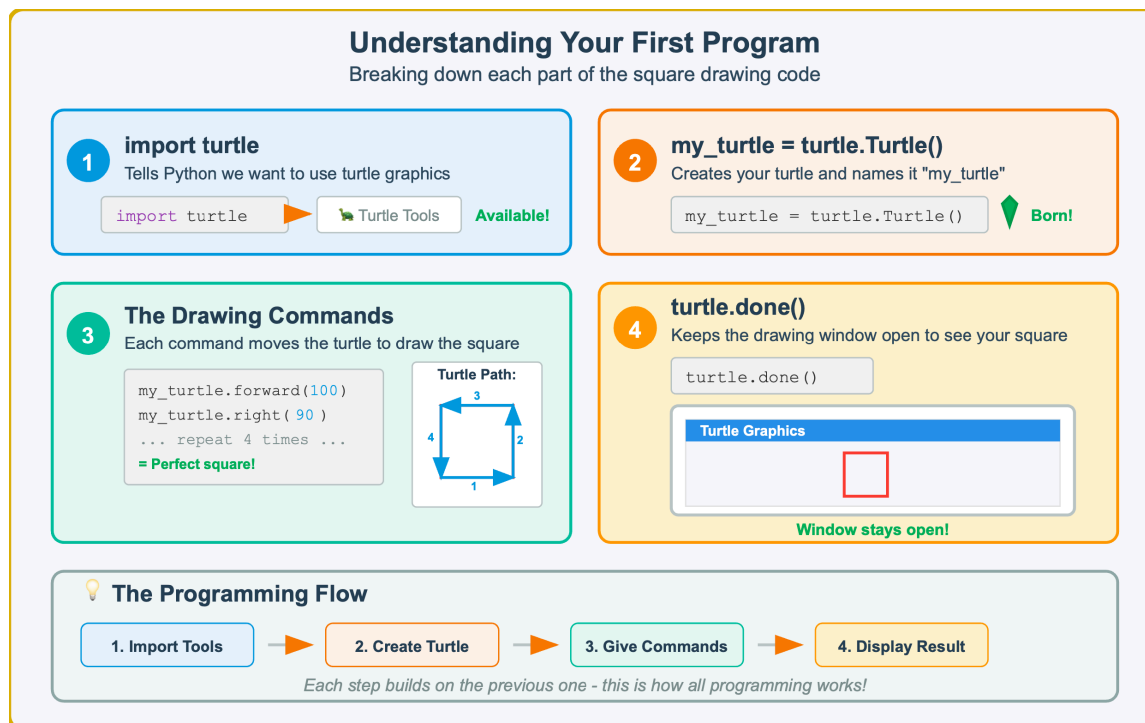


Figure 1.11 — Understanding Your First Program — Step—by—Step Code Breakdown. This detailed analysis shows how each part of your square drawing program works together, from importing tools to displaying results. Understanding this flow is fundamental to all programming.

Challenge: Create Your First Pattern

Use multiple squares to make something interesting!

 **Your Mission:**

Can you create a simple pattern by drawing multiple squares in different positions?
Try to make something that looks interesting to you!

Example Ideas to Inspire You

Side by Side Squares



Hint:

```
goto(50, 0)
# draw square
goto(100, 0)
# draw square
```

Overlapping Pattern



Hint:

```
goto(0, 0)
goto(10, 10)
goto(20, 20)
# etc...
```

Staircase Design



Hint:

```
x = x + 20
y = y + 15
goto(x, y)
# repeat
```

Starter Code Template

```
1 import turtle
2 my_turtle = turtle.Turtle()
3 # Draw your first square here
4 # Move to a new position
5 # my_turtle.goto(x, y)
6 # Draw your second square here
7 # Try more squares!
8 turtle.done()
9
```

Helpful Hints

- Moving Without Drawing**
Use `turtle.goto(x, y)` to move the turtle
- Experiment with Numbers**
Try different `x` and `y` values to see what happens
- Have Fun!**
Don't worry about perfection - just explore!

 **Remember: There's no wrong way to be creative - just have fun experimenting!**

Figure 1.12 — Challenge — Create Your First Pattern with Multiple Squares. This hands—on challenge encourages experimentation with `turtle.goto()` to create interesting patterns. Use the starter code and example ideas to build your own unique designs.